

Initial Project Planning Template

Date	03 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I can collect and clean the historical rainfall and flood datasets so that they are ready for exploratory analysis.	2	High	Lakshmi Lathish Pasupuleti	25 June 2026	25 June 2026
Sprint-1		USN-2	As a developer, I can perform exploratory data analysis on the rainfall and flood datasets to understand seasonal trends and correlations.	2	High	Lokasree Ranga Priya Duddu	25 June 2026	25 June 2026

Sprint-2	Model Training & Evaluation	USN-3	As a developer, I can train and compare Decision Tree, Random Forest, K-Nearest Neighbors and XGBoost classifiers on the flood dataset.	3	High	Lakshmi Lathish Pasupuleti	26 June 2026	26 June 2026
Sprint-2		USN-4	As a developer, I can evaluate model accuracy using a train/test split and select the best performing model for deployment.	2	High	Lokasree Ranga Priya Duddu	27 June 2026	27 June 2026
Sprint-2		USN-5	As a developer, I can save the trained model and scaler using joblib so the web app can load it at startup.	1	Medium	Lakshmi Lathish Pasupuleti	28 June 2026	28 June 2026
Sprint-3	Flask Web Application	USN-6	As a user, I can view the home page showing the project overview, selected model, and model accuracy.	1	Medium	Lokasree Ranga Priya Duddu	28 June 2026	28 June 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-3		USN-7	As a user, I can enter atmospheric readings and seasonal rainfall values on the prediction console form.	2	High	Lakshmi Lathish Pasupuleti	29 June 2026	29 June 2026
Sprint-3		USN-8	As a user, I can view the Flood Chance result page with the predicted probability and recommended next steps.	2	High	Lokasree Ranga Priya Duddu	29 June 2026	29 Jul 2026

Sprint-3		USN-9	As a user, I can view the No Flood Chance result page when the model predicts no flood risk.	1	Medium	Lakshmi Lathish Pasupuleti	30 June 2026	30 June 2026
Sprint-4	Deployment	USN-10	As a developer, I can containerize the application with Docker so it can be deployed to IBM Cloud Code Engine.	2	High	Lokasree Ranga Priya Duddu	01 July 2026	01 July 2026
Sprint-4		USN-11	As a developer, I can configure the Procfile and Cloud Foundry manifest for buildpack-based deployment.	1	Low	Lakshmi Lathish Pasupuleti	01 July 2026	01 July 2026